DUALITY AI OFFROAD SEMANTIC SCENE SEGMENTATION Startathon Hackathon 2026

– Segmentation Track

Team Name: Digital Disrupters
Team Members:

- Vansh Singh (25BCE10723)
- Chinmay Kumar (25BCE10865)
- Ravi Prakash Rai (25BAI11199)
- Sammireddy Varshini Reddy (25BAI11554)

Role Distribution: AI Engineering + Documentation

----

1. PROBLEM STATEMENT

The objective of this project is to train a multi-class semantic segmentation model using a synthetic desert dataset generated from Duality AI's Falcon digital twin platform.

The model must classify each pixel in an image into one of the following 10 classes:

- Trees
- Lush Bushes
- Dry Grass
- Dry Bushes
- Ground Clutter
- Flowers
- Logs
- Rocks
- Landscape
- Sky

Unlike image classification, semantic segmentation requires pixel-wise prediction, making it a dense spatial learning task.

----

2. METHODOLOGY

2.1 Training Environment

Framework: PyTorch Environment: Anaconda (EDU environment) GPU: NVIDIA GTX 1650 (4GB VRAM) Optimizer: Adam Learning Rate: 1e-4 Batch Size: 2

Loss Function: Cross Entropy Loss

Due to GPU memory constraints (4GB VRAM), a small batch size was used to ensure stable training without memory overflow.

A DINOv2 Vision Transformer backbone was used for feature extraction. A lightweight ConvNeXt-style segmentation head was trained on top of the frozen backbone embeddings.

2.2 Training Workflow

- Model trained using train.py on provided training dataset.
- Validation performed on separate validation set.
- Final evaluation conducted on unseen test images.
- Strict separation maintained between training and test data.

2.3 Experimental Setup

Baseline Model: Epochs: 10 Validation IoU: 0.2935 Test IoU: 0.2275

Extended Training: Epochs: 25 Validation IoU: 0.3099 Test IoU: 0.2272

The extended training was conducted to improve spatial feature learning and generalization.

----

3. EVALUATION METRIC

Model performance was evaluated using Mean Intersection over Union (Mean IoU).

IoU = TP / (TP + FP + FN)

Mean IoU measures overlap between predicted segmentation masks and ground truth masks across all classes. It penalizes both false positives and false negatives and is more suitable for segmentation tasks than raw accuracy.

----

4. RESULTS AND ANALYSIS

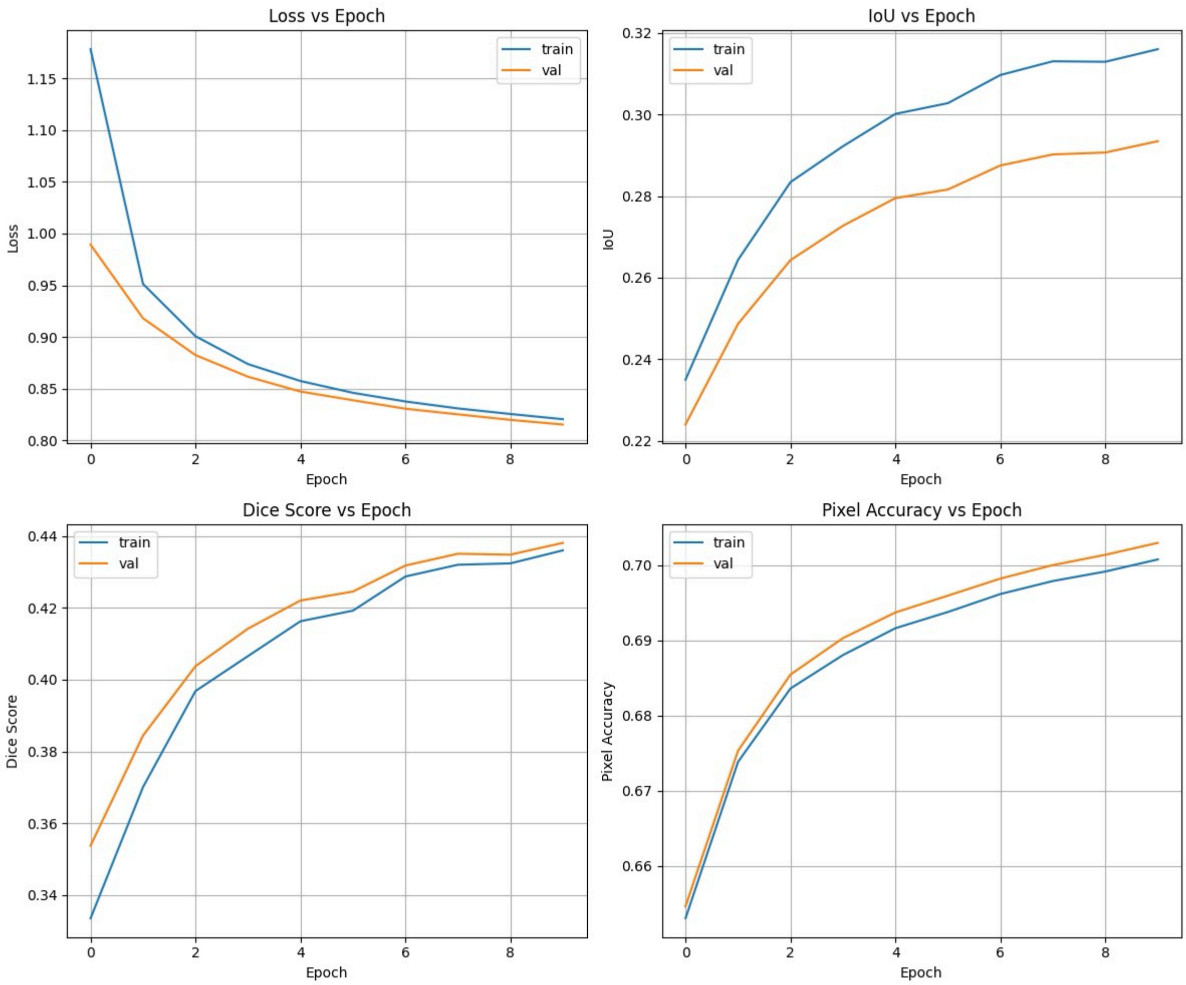
Observations:

- Validation IoU increased steadily during extended training.
- Training loss decreased consistently.
- Model demonstrated generalization on unseen desert test images.

First Training (10 epochs):-

```
Final evaluation results:
  Final Val Loss:      0.8154
  Final Val IoU:       0.2935
  Final Val Dice:      0.4381
  Final Val Accuracy:  0.7030

Training complete!
```



First Test:-

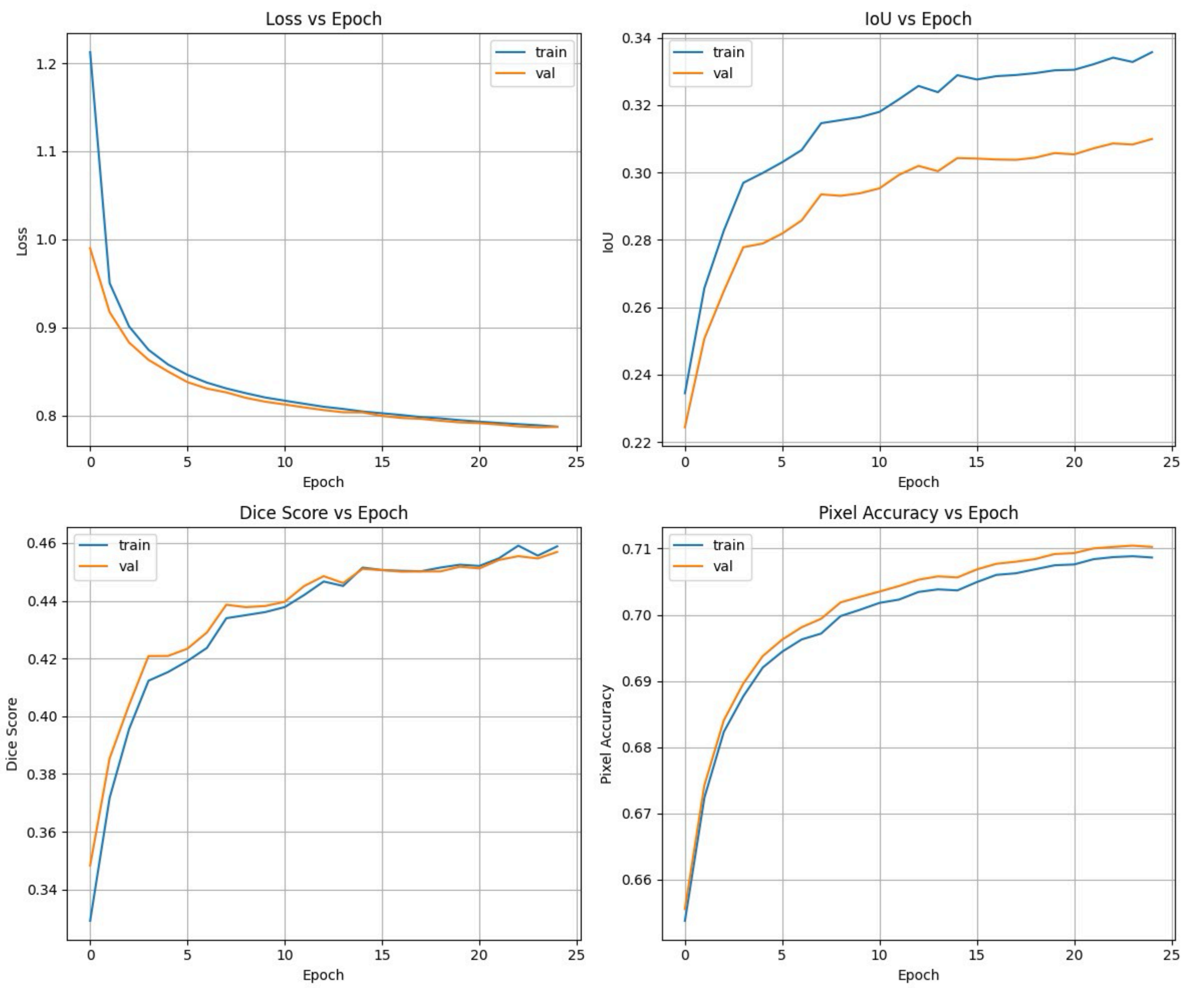
```
EVALUATION RESULTS
==================================================
Mean IoU:          0.2275
==================================================

Per-Class IoU:
----------------------------------------
  Background          : 0.0000
  Trees               : 0.0686
  Lush Bushes         : 0.0149
  Dry Grass           : 0.1475
  Dry Bushes          : 0.0540
  Ground Clutter      : 0.0527
  Logs                : 0.0077
  Rocks               : 0.2321
  Landscape           : 0.5282
  Sky                 : 0.9389
```

Second Training(25 epochs):-

```
Final evaluation results:
  Final Val Loss:     0.7871
  Final Val IoU:      0.3099
  Final Val Dice:     0.4569
  Final Val Accuracy: 0.7102

Training complete!
```



Second Test:-

```
EVALUATION RESULTS
==================================================
Mean IoU:          0.2272
==================================================

Per-Class IoU:
----------------------------------------
  Background          : 0.0000
  Trees               : 0.0876
  Lush Bushes         : 0.0151
  Dry Grass           : 0.1538
  Dry Bushes          : 0.0604
  Ground Clutter      : 0.0515
  Logs                : 0.0117
  Rocks               : 0.2049
  Landscape           : 0.5046
  Sky                 : 0.9531
```

The decrease in IoU on the unseen test dataset indicates domain shift between training and test environments. While large classes such as Sky and Landscape maintained strong performance, smaller vegetation classes experienced reduced IoU, suggesting sensitivity to texture variation.

The baseline model (10 epochs) achieved a validation IoU of 0.2935, while extended training (25 epochs) improved validation IoU to 0.3099. This demonstrates that additional epochs improved spatial feature learning and convergence stability.

---

5. FAILURE CASE ANALYSIS

Observed challenges:

- Confusion between Dry Grass and Dry Bushes due to visual similarity.
- Logs occasionally misclassified as Ground Clutter.
- Small objects (Flowers) harder to detect due to scale imbalance.

These errors indicate texture-based similarity challenges in desert environments.

---

6. CHALLENGES FACED

- Limited GPU Memory (4GB): Required reducing batch size to 2.
- Long Training Time: Pixel-wise segmentation significantly increases computational load.
- Class Similarity: Vegetation classes share similar color and texture features.

---

7. CONCLUSION

A multi-class semantic segmentation model was successfully trained using synthetic desert data.

Extended training improved generalization performance and IoU score. The model demonstrates stable convergence, effective large-scale spatial understanding, and reasonable robustness under hardware constraints. Performance analysis indicates that further gains may be achieved through class-weighted loss and augmentation strategies.

---

8. FUTURE WORK

- Implement class-weighted loss to handle imbalance.
- Apply data augmentation techniques.
- Introduce learning rate scheduling.

- Use mixed precision training for speed optimization.
- Explore domain adaptation techniques.

-END OF REPORT